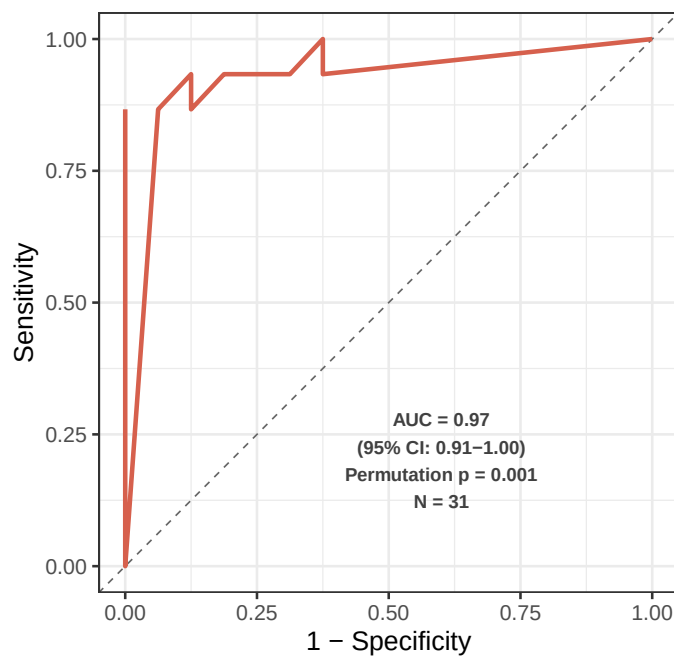
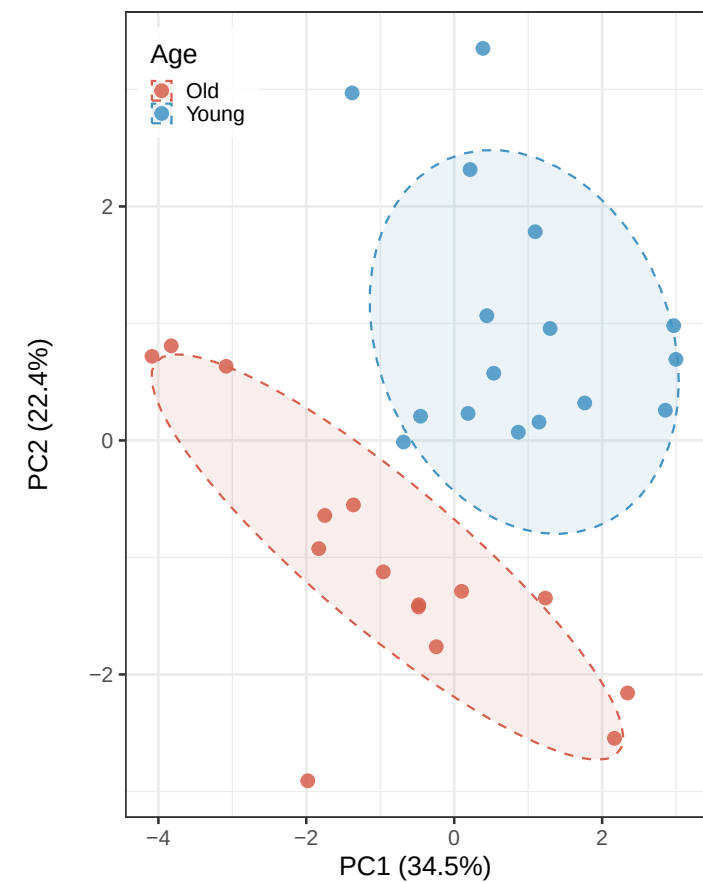


Translational Utility of WGCNA Modules

Module eigengenes discriminate age groups, associate with training response, and link to phenotypic change.

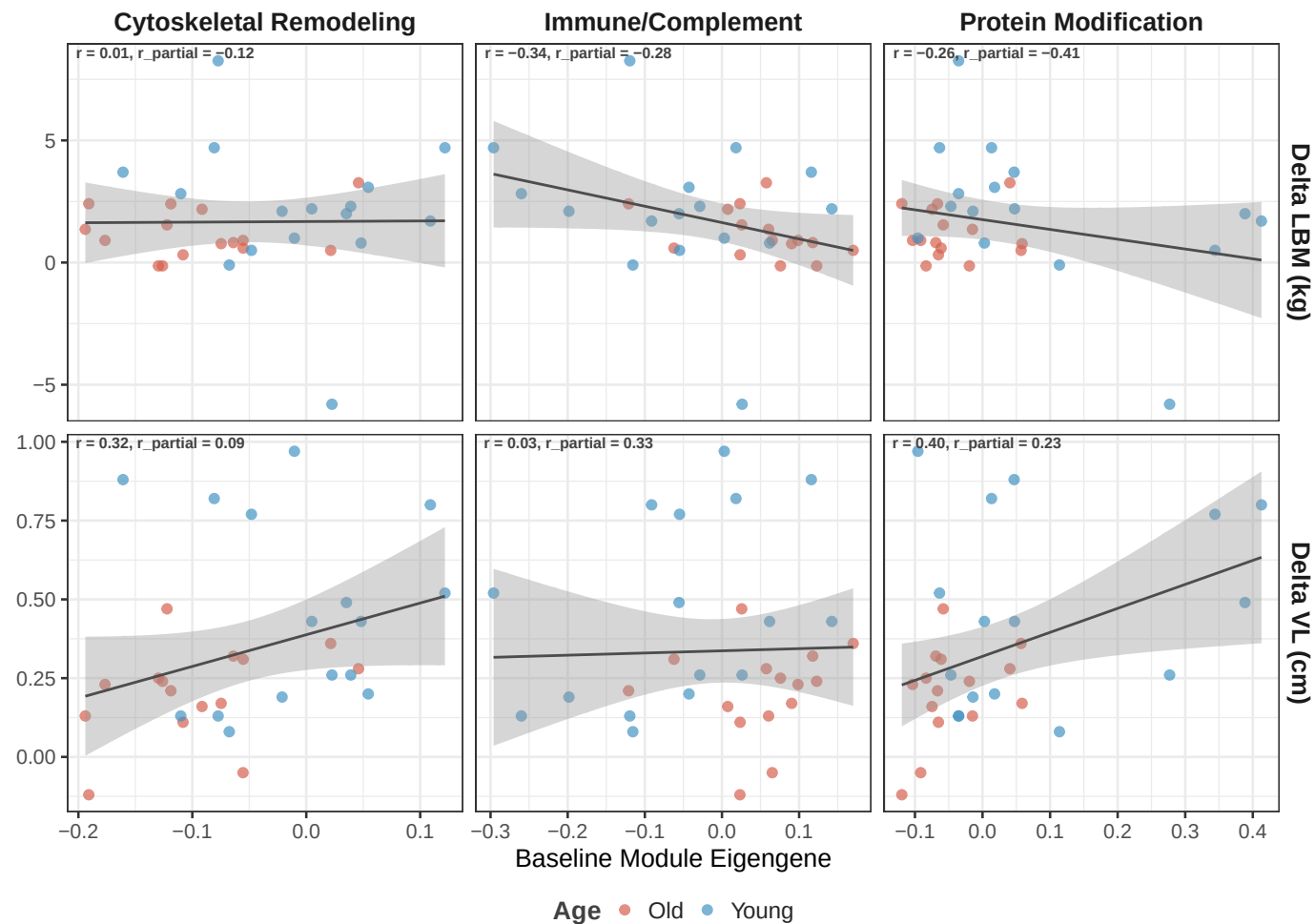
A Age Discrimination via Module Eigengenes

PCA of 10 baseline eigengenes | nested LOOCV (top 5 per fold)



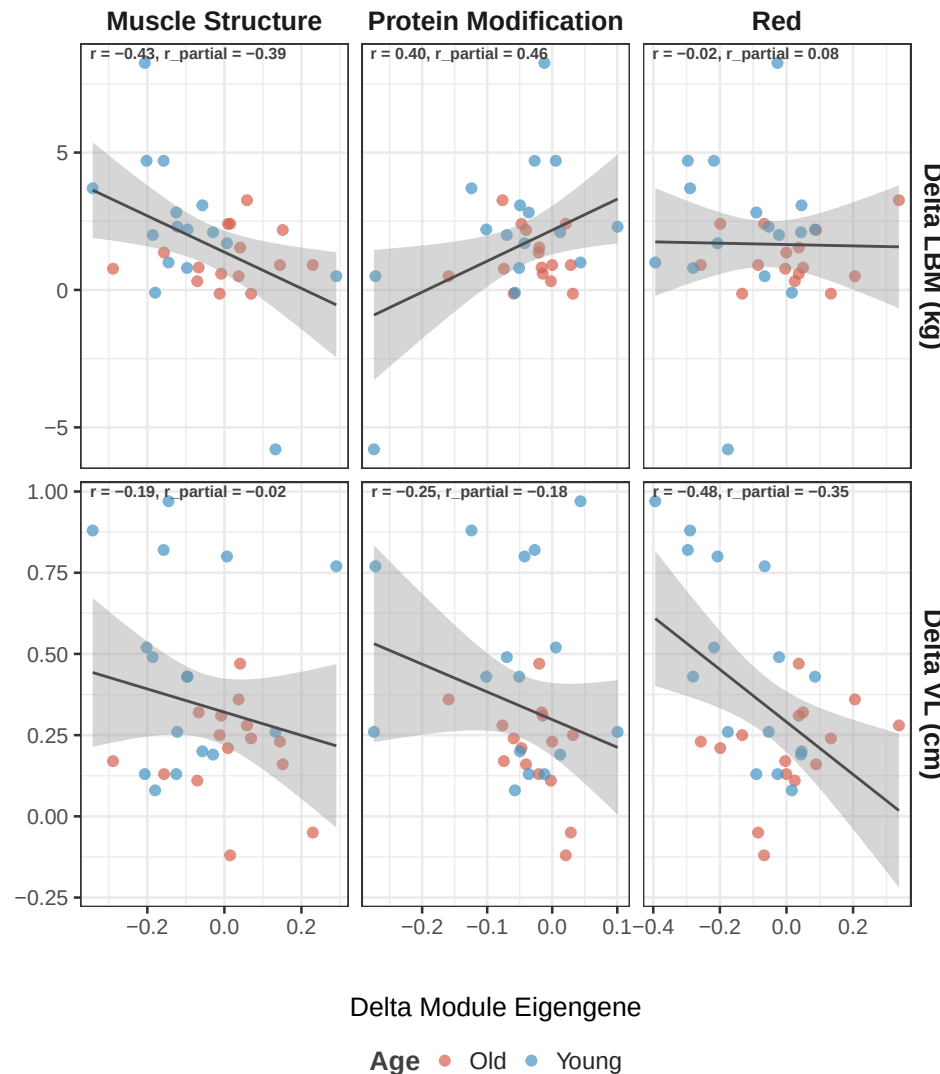
B Baseline Eigengene vs Training Response

Baseline module eigengene (Pre) vs change in phenotype | No facet reaches $p_{adj} < 0.05$



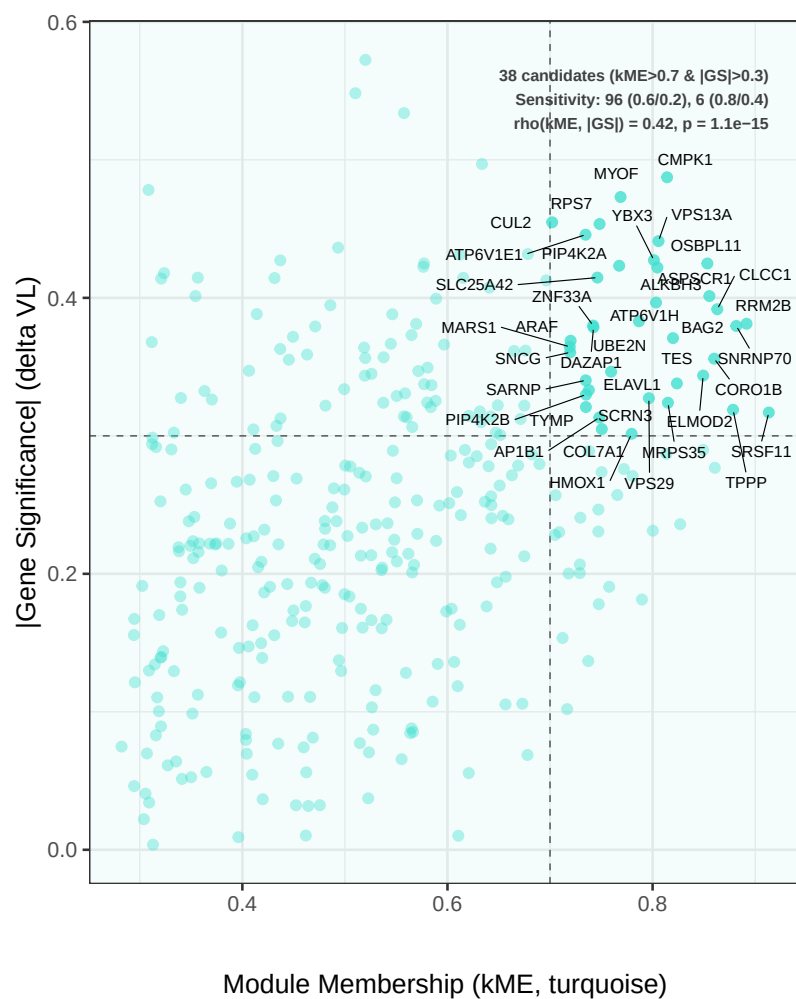
C Training-Induced Module Changes Track Phenotype

Delta eigengene (Post-Pre) vs delta phenotype



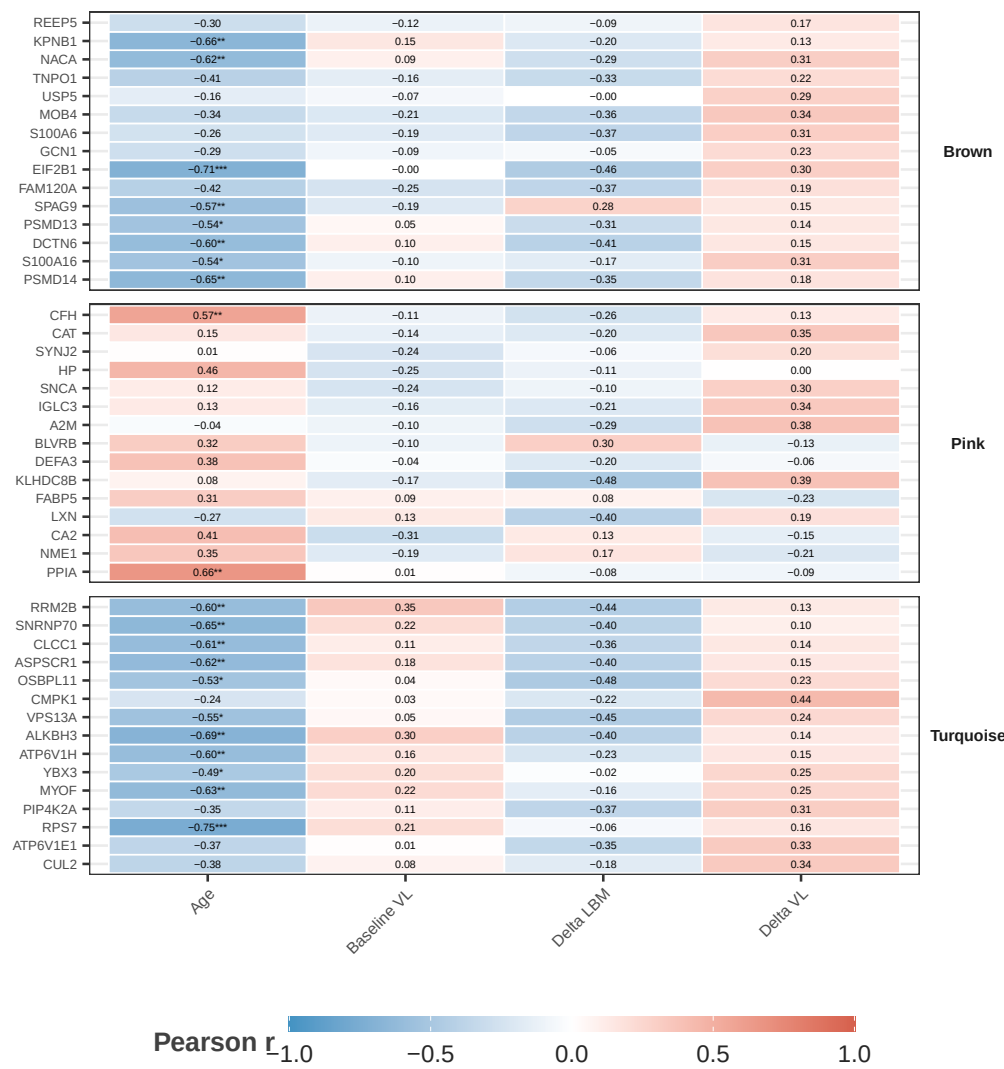
D Hub Proteins (Turquoise Module)

Module Membership vs |Gene Significance| for delta VL



E Protein-Phenotype Associations

Top hub proteins (baseline vs traits) | BH-corrected across 180 tests | Delta traits across



Age: Young Old Top Modules: Turquoise Pink Brown Pearson r: -1 0 +1 Phenotype: Delta VL